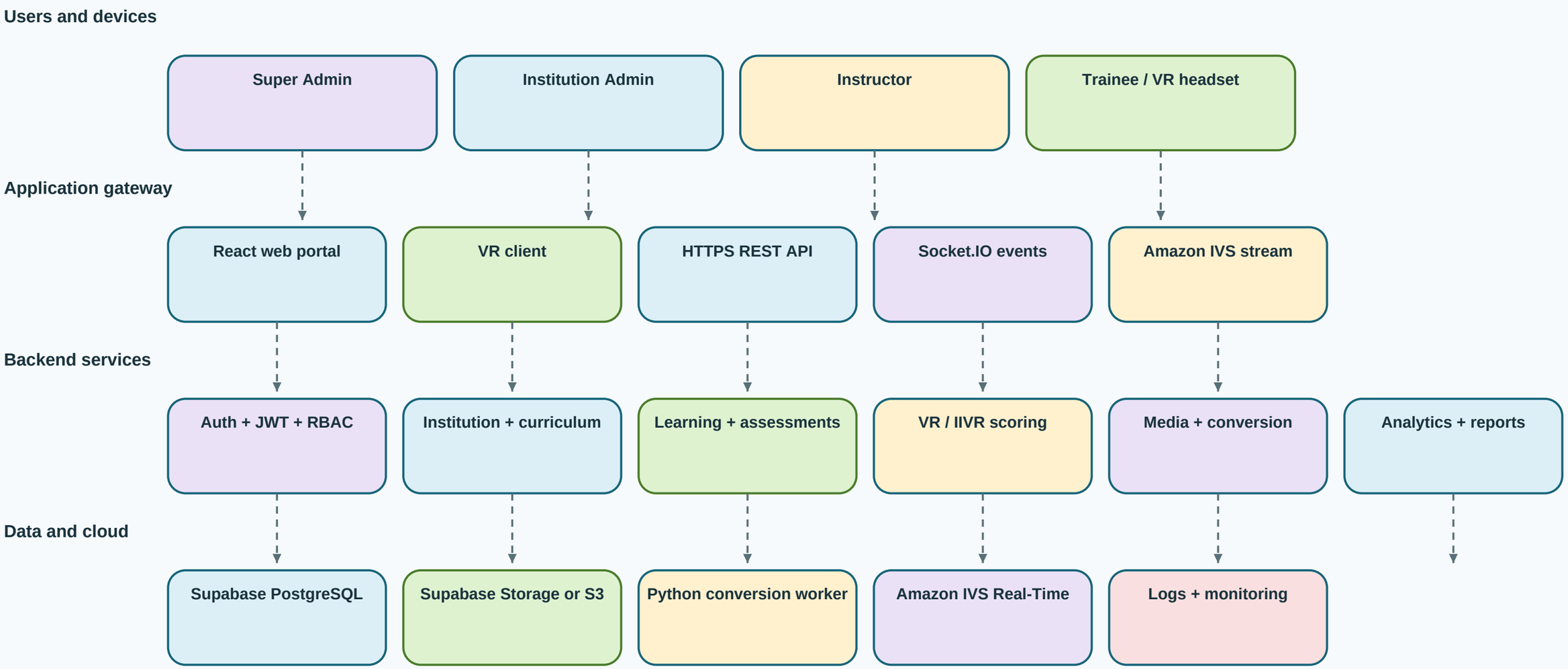
Project ANU - Centralized technical architecture

All roles use the same secured platform services and persistence layers



Primary request path: user or headset -> REST API -> authorized backend service -> database or object storage.

Realtime path: headset -> Amazon IVS -> instructor browser; activity/status events -> Socket.IO -> dashboards.

Every protected request carries the authenticated user, role, institution scope and session context.

Project ANU - Role-based technical flows

The four roles share the platform but have different entry points and permissions

99 Super Admin

Login -> role gate -> global dashboard
Manage institutions, admins, curriculum, certifications, global volumes, mappings, content access and audit reports.

Technical route

- POST /login
- JWT role 99
- Global service calls
- PostgreSQL writes
- Object storage admin
- Analytics / audit

101 Institution Admin

Login -> institution scope -> operations dashboard
Create batches, assign instructors, enroll trainees, approve volumes, schedule learning and monitor streams.

Technical route

- POST /login
- JWT role 101
- Institution scope
- Batch + enrollment
- Approval + streams
- Reports / remediation

102 Instructor

Login -> assigned batch scope -> instructor dashboard
Upload volume and recordings, review trainees, supervise live sessions and recommend reattempts.

Technical route

- POST /login
- JWT role 102
- Batch scope
- Upload + review
- IVS viewer token
- Feedback / reattempt

103 Trainee

Login from browser or headset -> assigned learning
Study resources, complete MindSpark, run VR practice/tests, submit results and view feedback.

Technical route

- POST /login
- JWT role 103
- Assigned content
- Activity submission
- VR session / IVS
- Score + progress

RBAC is enforced at login, protected routes, controller/service authorization and institution/batch data scope.

Project ANU - Complete platform lifecycle

From platform setup to learning, evaluation, analytics and remediation

1. Configure

Super Admin
Institution, admins, curriculum and resources

2. Organize

Institution Admin
Batch, instructor, trainees and schedules

3. Prepare assets

Instructor / Admin
Upload, approve, convert, calibrate and map volumes

4. Assign

Admin
Publish course and targeted learning to a batch

5. Learn

Trainee
Theory, video, MindSpark and challenges

6. Simulate

Trainee + headset
Load volume, scan, annotate, measure and submit

7. Review

Instructor / Admin
Scores, errors, recordings, streams and feedback

8. Improve

Analytics engine
Competency, benchmark, recommendation and reattempt loop

Volume

Raw .vol -> object storage -> worker -> .nii/.nii.gz -> mapped scenario

Session

Headset -> REST API -> session record -> result submission -> score

Live monitor

Headset publisher -> IVS stage -> instructor viewer

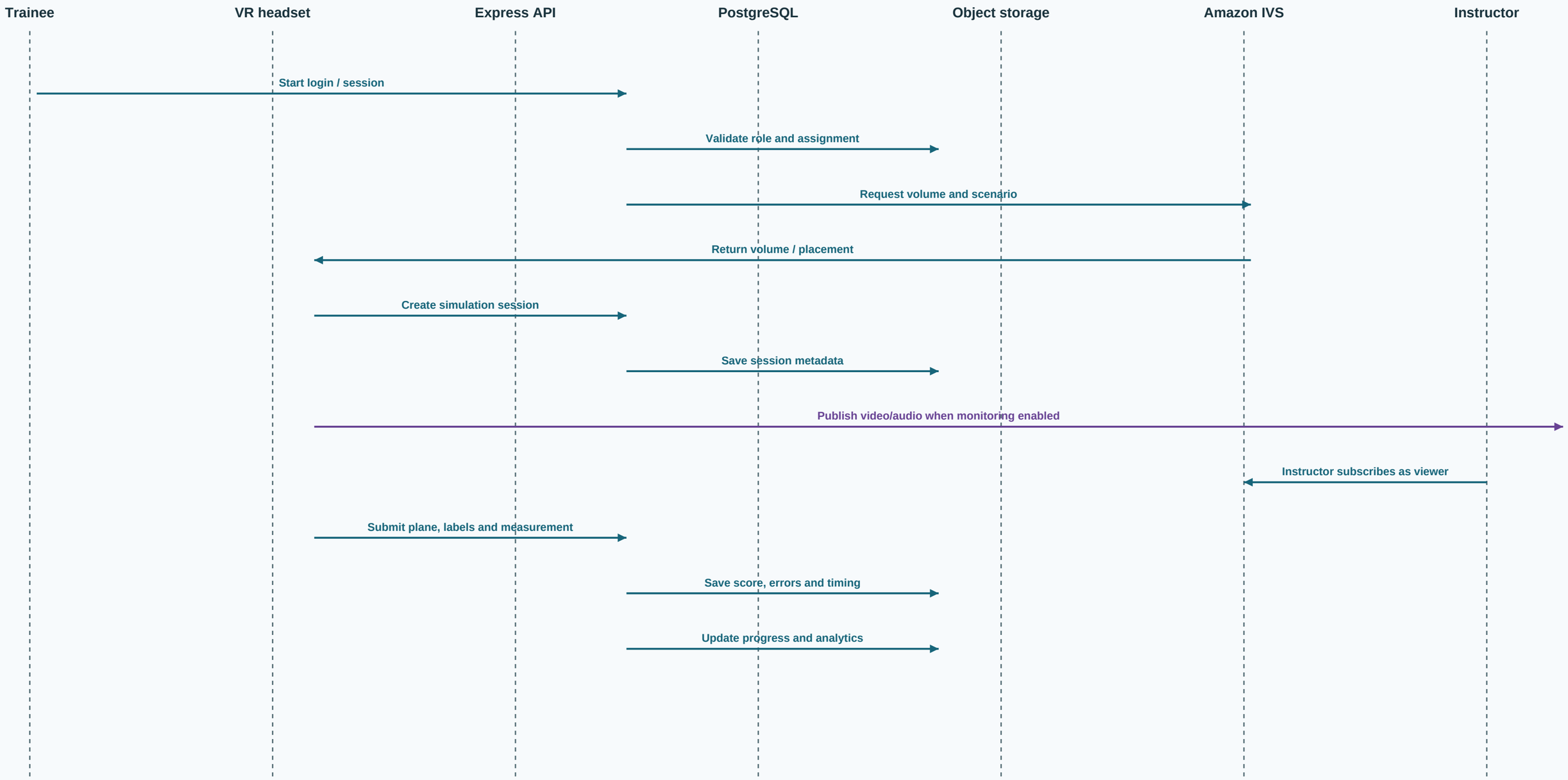
Data loop

PostgreSQL + object storage -> analytics -> role dashboards

Decision points: volume approval, account authorization, live supervision, task completion and remediation requirement.

Project ANU - VR, headset, LMS and live communication flow

Technical sequence for an authenticated simulation with optional instructor monitoring



Technical result

Headset receives feedback; instructor sees live or completed-session status; dashboards read persisted results.

Project ANU - Storage, communication and cloud cost map

Cost must be measured separately for database, storage, conversion, streaming, transfer and monitoring

Flow point	Data / service	Cost item	Main driver
Login / LMS API	Auth, courses, dashboards	API compute, requests, logs	Requests and execution time
PostgreSQL	Users, batches, scores, sessions	Supabase plan and database usage	Rows, queries, connections
Raw volumes	.vol ultrasound files	Object storage capacity and PUTs	File size and upload count
Conversion	.vol -> .nii/.nii.gz	Worker CPU, memory and runtime	Conversion duration and size
VR downloads	NIfTI and scenario assets	Storage GETs and data transfer	Session count and volume size
Results/media	Images, recordings, reports	Storage and database writes	Attempts and retention
Live supervision	WebRTC video/audio	Amazon IVS stage and delivery	Stream minutes, viewers
Realtime status	Socket.IO activity events	Realtime compute and logs	Connected users and events
Monitoring	API, worker and IVS logs	Cloud logs and metrics	Volume and retention

Storage locations

PostgreSQL / Supabase: users, roles, batches, courses, sessions, scores and logs.
Object storage: raw volumes, NIfTI, images, recordings and reports.

Communication paths

HTTPS REST: browser or headset to Express API.
WebRTC / IVS: headset stream to instructor. Socket.IO: activity and status events.

Pricing assumptions needed

Active users, sessions per month, average volume size, retention, concurrent viewers, AWS/Supabase region and plan.

Use the cost categories above to attach actual provider rates and monthly usage. Do not combine IVS or conversion cost into a generic API estimate.